Closing the Gap: Cleft Training in Pediatric Dentistry Residency Programs

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Summary

In this study, we analyzed data within residents and directors, then compared the groups' answer selections, informing the difference that various factors can have on a resident's knowledge and experience in their program, and if the directors have a different perception of what residents face during residency. Out of 10 tests within the residents group, we found 4 to be significant. When cross-comparing the groups on each common question, we found 8/23 comparisons to be significant, 5 after adjustment. This means that at least 5 questions saw significant evidence of being answered differently for residents and directors. All tests will be reported in the significant findings section, and their significance before and after adjustment will be noted. It should be noted that some questions were requested specifically from the program director survey, but those comparisons were covered by comparing each resident/director combination in the cross-comparison section, and their significance will be reported in the significant findings section as well.

Methods

For our methodology, we conducted Fisher exact tests, which can calculate exact probabilities to see if there's a non-random association between groups. A low p-value (after adjustment) translates to having significant evidence for there existing some association between questions.

For cross referencing 2x11x12, we had to run a regression model, where there were multiple questions being compared with the response variable, which in this case was whether a resident was likely to continue their practices after residency, and this model was also found to be significant.

For questions that had multiple parts, we had to split them into individual questions (e.g. 11a, 11b, 11c…) to see if there were specific responses that were influenced by the grouping variable in question, which was the role (being a resident/director) in the case of cross-comparison,

or information like program, experience, etc. In practice, this saw us, for example, analyzing whether directors perceive residents having knowledge on related, but separate, types of care. This example came from data from question 11 on the director's survey to illustrate the complexity of certain questions.

Then, to visualize the difference, we created side-by-side bar charts which plot the distribution of question responses versus the grouping variable. For the specific 2x11x12 question, a visualization of the regression model was created to show the probability of a resident continuing in their program after finishing residency. For resident/director comparison, this shows side-by-side plots of the residents' response to a question versus the directors'. Noticeable/interesting plots and findings were placed in the "significant findings" section of this writeup and all plots, significant or not, are in the appendix.

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.2.0
-- Conflicts ---------------------------------------- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco

Attaching package: 'zoo'


The following objects are masked from 'package:base':

    as.Date, as.Date.numeric
```

Results Tables

Residents Results

	Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
1	Q8a	Q2	0.001	TRUE	fisher	0.0033	TRUE
2	Q8b	Q2	0.001	TRUE	fisher	0.0033	TRUE
3	Q8c	Q2	5e-04	TRUE	fisher	0.0033	TRUE
4	Q8d	Q2	0.0035	TRUE	fisher	0.0088	TRUE
5	Q12	Q2	0.1194	FALSE	fisher	0.2388	FALSE

6	Q7a	Q2	0.8576	FALSE	fisher	0.8576	FALSE
7	Q7b	Q2	0.7041	FALSE	fisher	0.8576	FALSE
8	Q7c	Q2	0.7741	FALSE	fisher	0.8576	FALSE
9	Q7d	Q2	0.5147	FALSE	fisher	0.8576	FALSE
10	Q11	Q2	0.6427	FALSE	fisher	0.8576	FALSE

Cross-Comparison Results

	Question1	Question2	P_value	Significant	Test	Adj_P	Sig_Adj
1	Q8c	role	0.0015	TRUE	fisher	0.0345	TRUE
2	Q6	role	0.0045	TRUE	fisher	0.0431	TRUE
3	Q7b	role	0.007	TRUE	fisher	0.0431	TRUE
4	Q8a	role	0.0075	TRUE	fisher	0.0431	TRUE
5	Q8b	role	0.0145	TRUE	fisher	0.0667	FALSE
6	Q10a	role	0.031	TRUE	fisher	0.1188	FALSE
7	Q7d	role	0.043	TRUE	fisher	0.1351	FALSE
8	Q9	role	0.047	TRUE	fisher	0.1351	FALSE
9	Q7c	role	0.0675	FALSE	fisher	0.1725	FALSE
10	Q7a	role	0.087	FALSE	fisher	0.1861	FALSE
11	Q13	role	0.089	FALSE	fisher	0.1861	FALSE
12	Q5	role	0.1062	FALSE	fisher	0.2036	FALSE
13	Q8d	role	0.1289	FALSE	fisher	0.2281	FALSE
14	Q4	role	0.1619	FALSE	fisher	0.2660	FALSE
15	Q10e	role	0.2949	FALSE	fisher	0.4522	FALSE
16	Q10c	role	0.4193	FALSE	fisher	0.6027	FALSE
17	Q10b	role	0.5012	FALSE	fisher	0.6404	FALSE
18	Q12	role	0.4878	FALSE	fisher	0.6404	FALSE
19	Q10d	role	0.5847	FALSE	fisher	0.7078	FALSE
20	Q2	role	0.8311	FALSE	fisher	0.9558	FALSE
21	Q15	role	0.9025	FALSE	fisher	0.9885	FALSE
22	Q3	role	1	FALSE	fisher	1.0000	FALSE
23	Q14	role	1	FALSE	fisher	1.0000	FALSE

Significant Findings

Residents Significance

The following question combinations were found significant between questions in the residents survey. Their contexts and plots will be noted below:

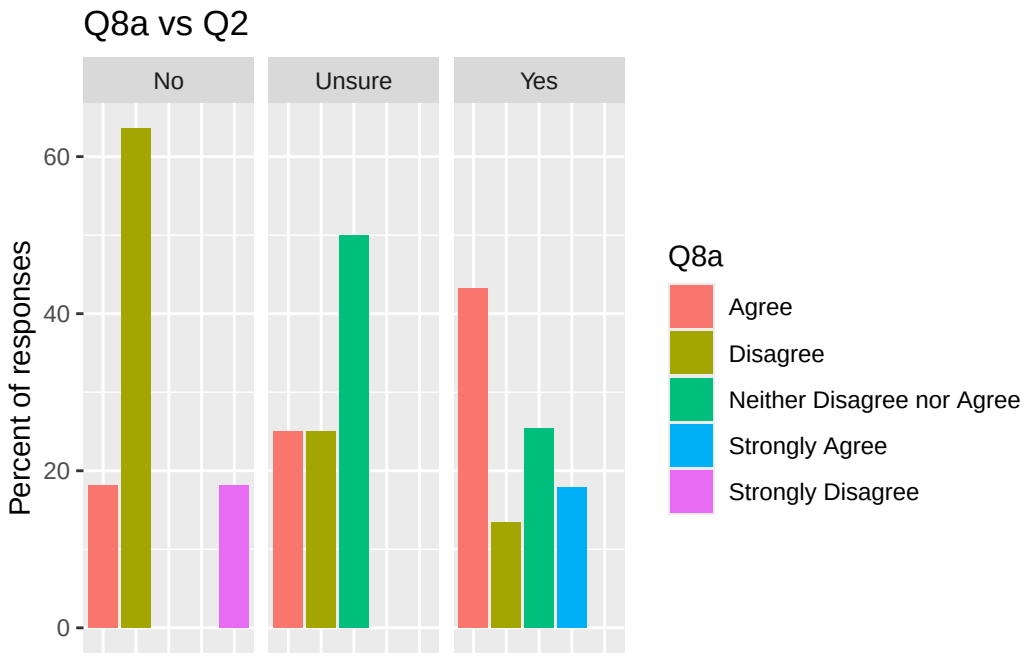
Q8a/b/c/d & 2: Feel prepared to:

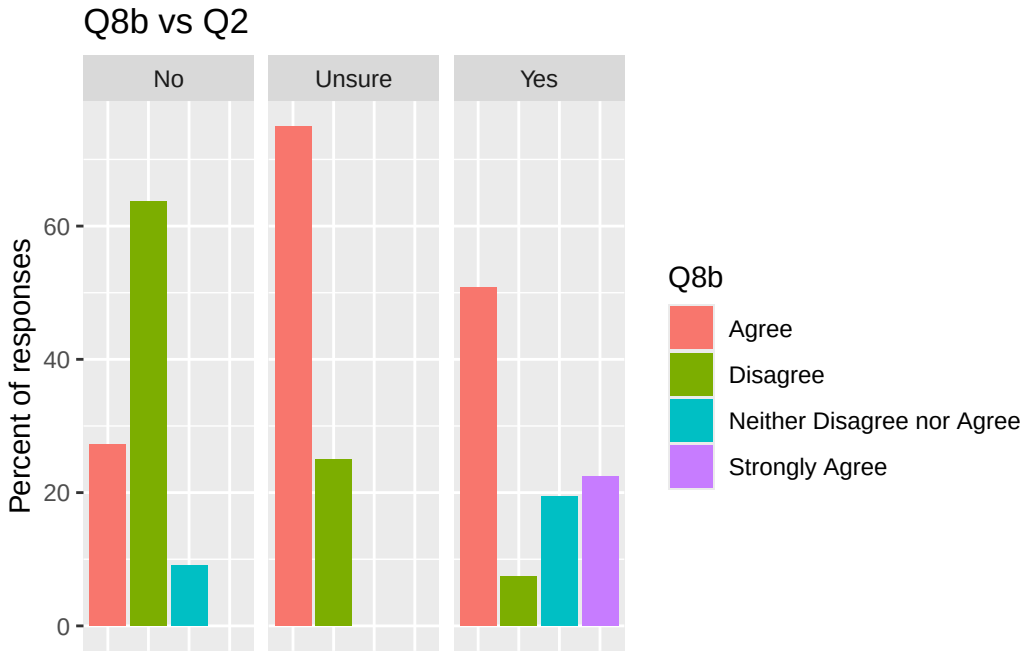
a) provide appropriate care for a patient with a cleft or craniofacial...

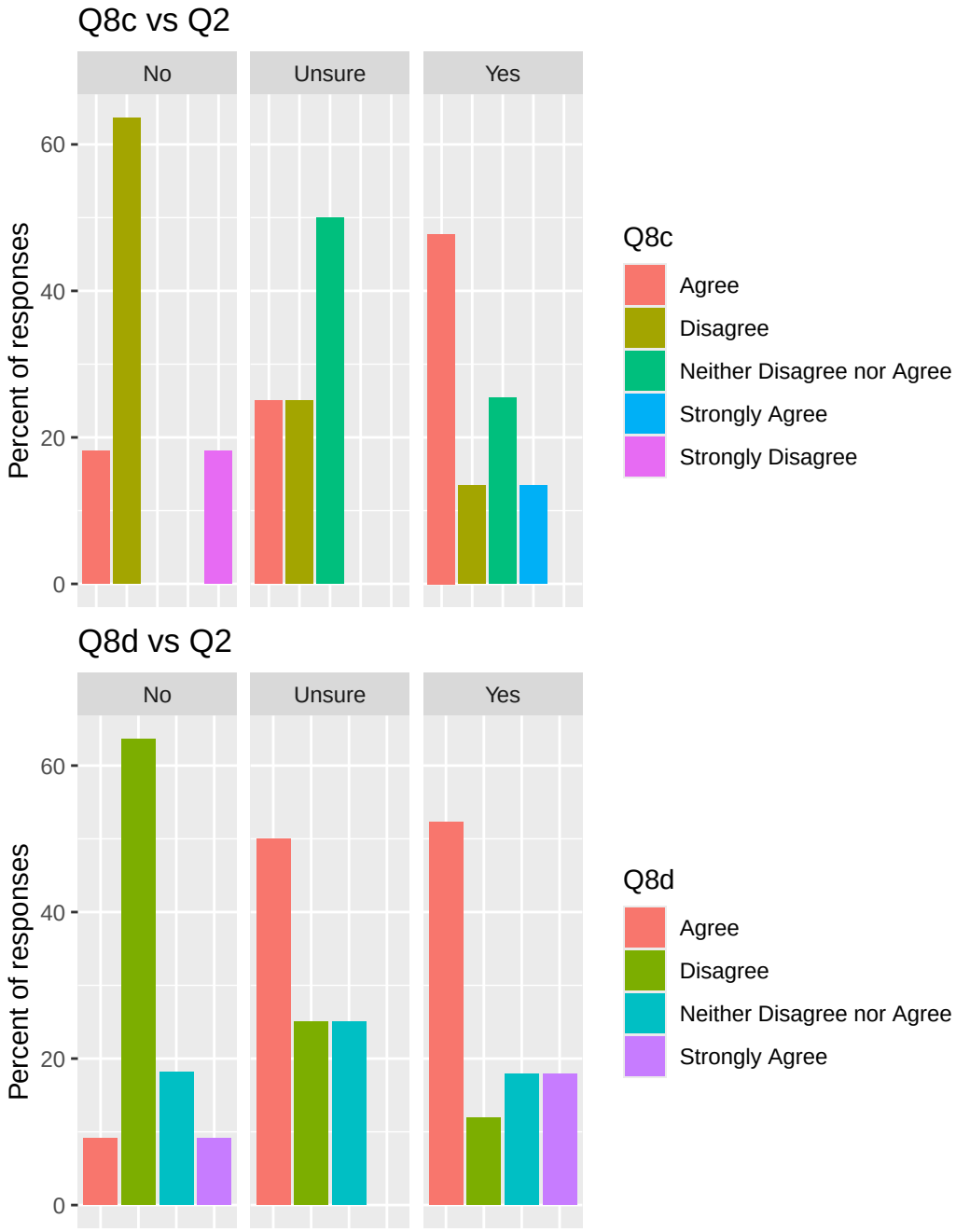
b) identify and collaborate with the appropriate specialists as part of a cleft...

c) manage pre-surgical and post-surgical dental care for patients with cleft or...

d) manage behavioral and psychosocial considerations when providing care to patients...







Q2x11x12: Association with cleft center/desire to be involved in cleft care post-residency/type of institution

We see from this model summary that we have one parameter that is particularly significant (being at a hospital-based program), but in total, the model predicts desire to be involved in cleft care more significantly than a model without any predictors (essentially, it's much better

	(1)
(Intercept)	1.059
	(0.865)
cleft_centerYes	0.713
	(0.758)
programHospital-based	−1.635
	(0.623)
programUniversity	0.236
	(1.290)
Num.Obs.	77
AIC	97.2
BIC	106.6
Log.Lik.	−44.598
F	2.950
RMSE	0.45

than random chance). Our coefficient for hospital-based programs is -1.635, which translates to saying that if a resident is in a hospital-based program, the odds of them NOT wanting to continue into cleft care is ~5 times the odds of a resident in a hybrid program.

Cross-Comparison Significance

Many questions were found to be significant between residents and directors, and below are all the questions that they responded significantly differently to:

Q6: Instruction in all aspects of cleft reconstruction

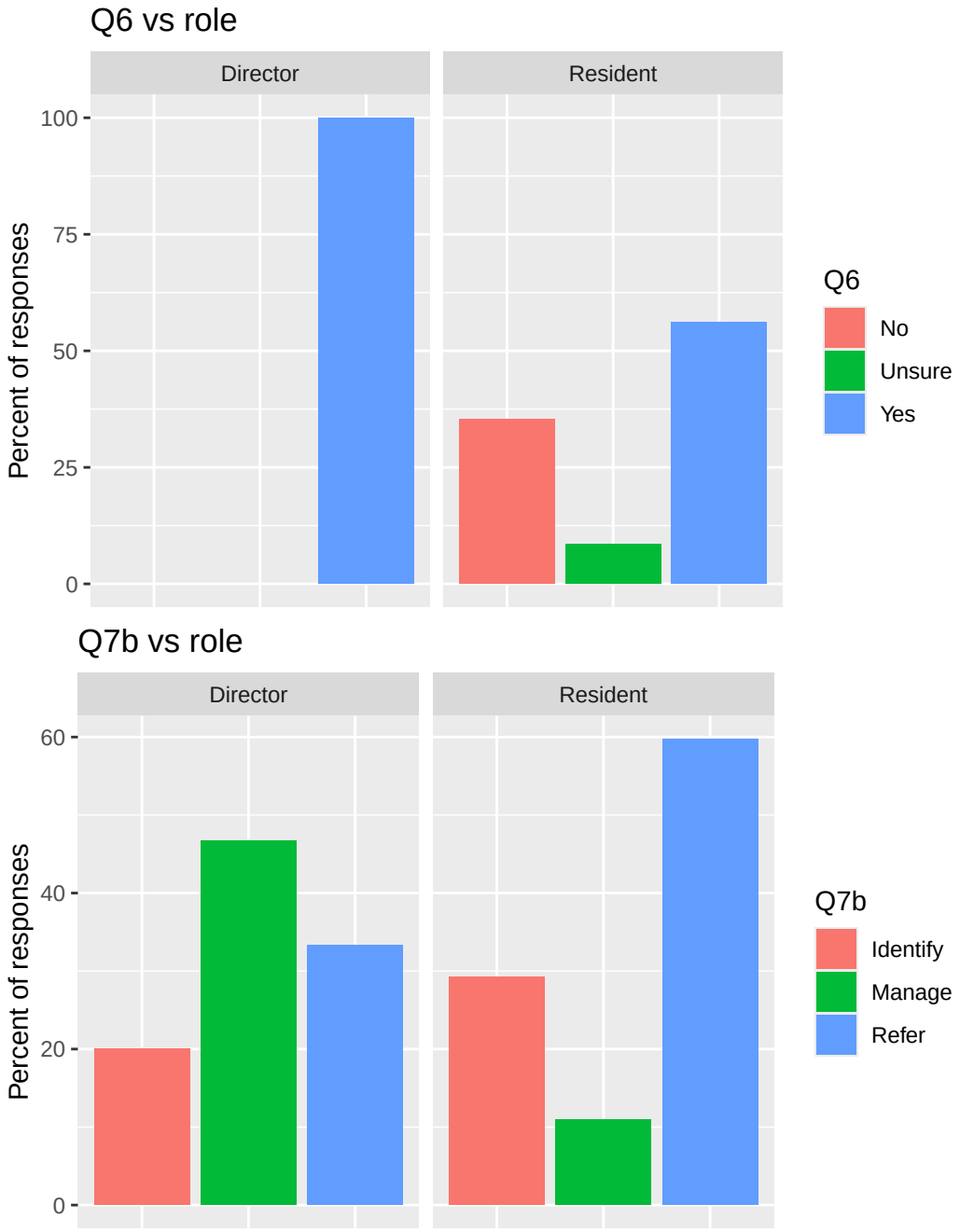
Q7b: Level of education in the appropriate timing of palatal closure

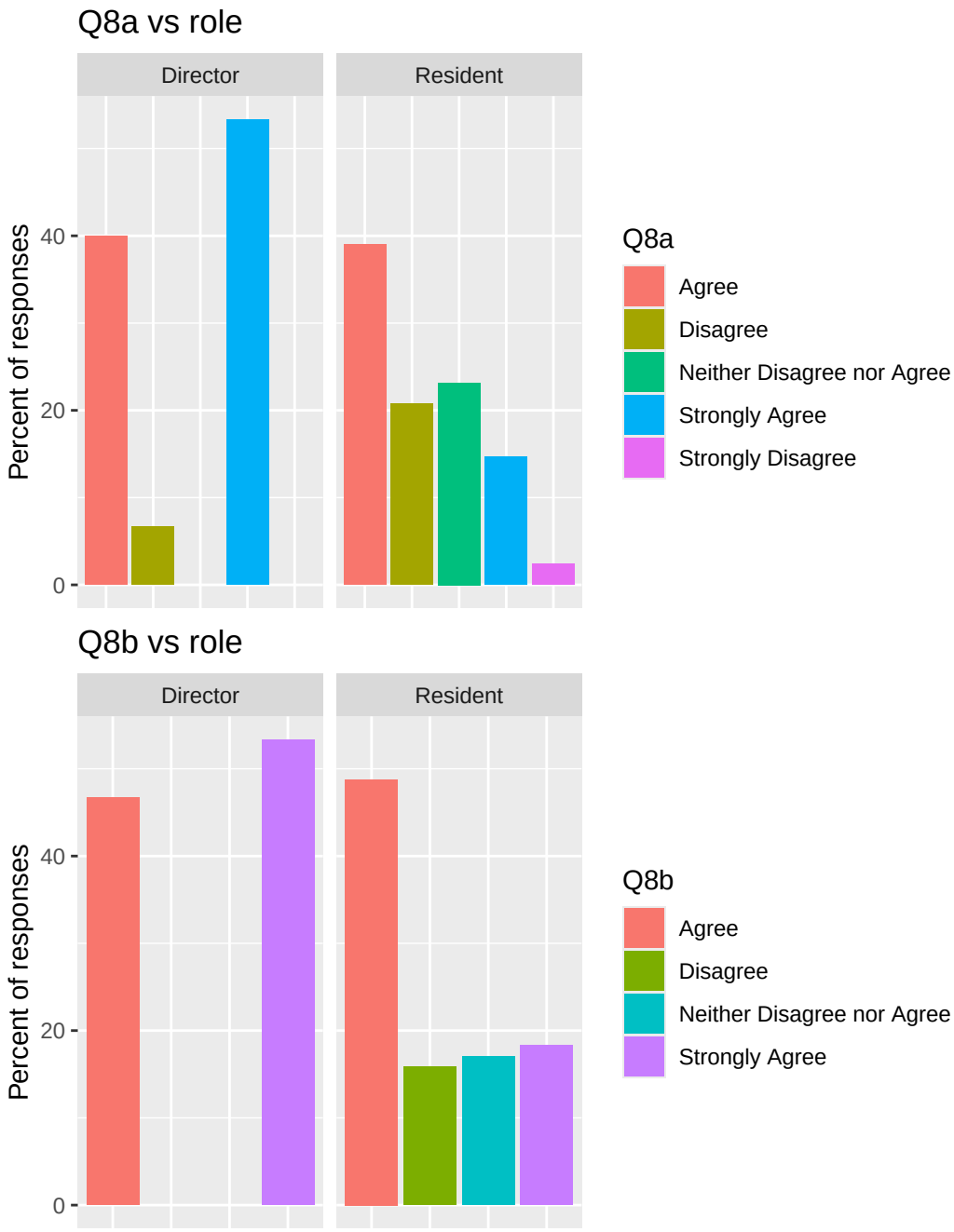
Q8a/b/c: Feel prepared to:

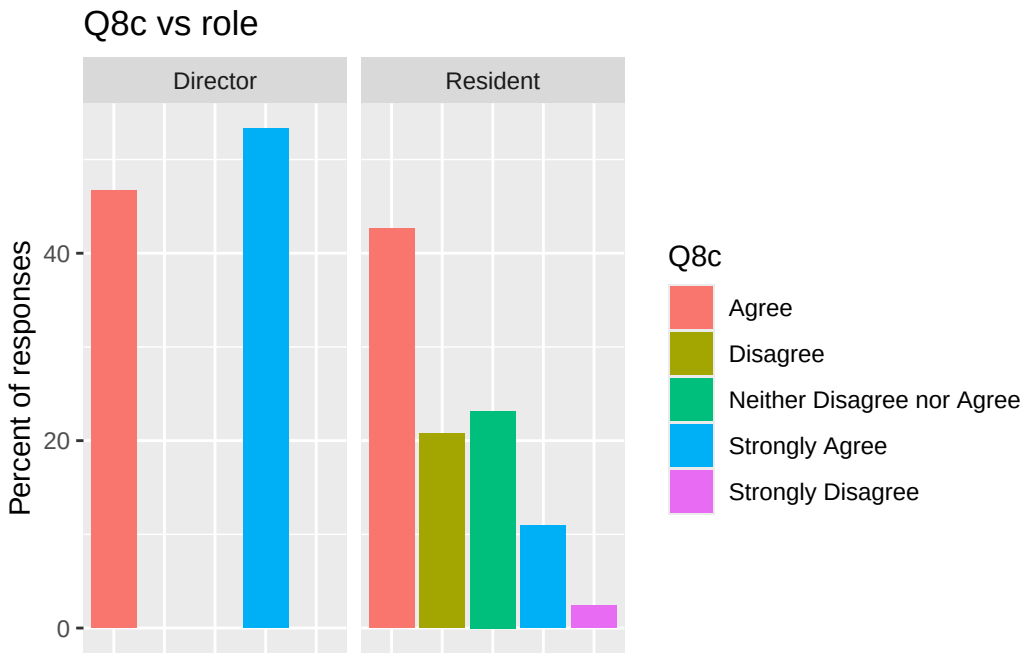
a) provide appropriate care for a patient with a cleft or craniofacial…

b) identify and collaborate with the appropriate specialists as part of a cleft…

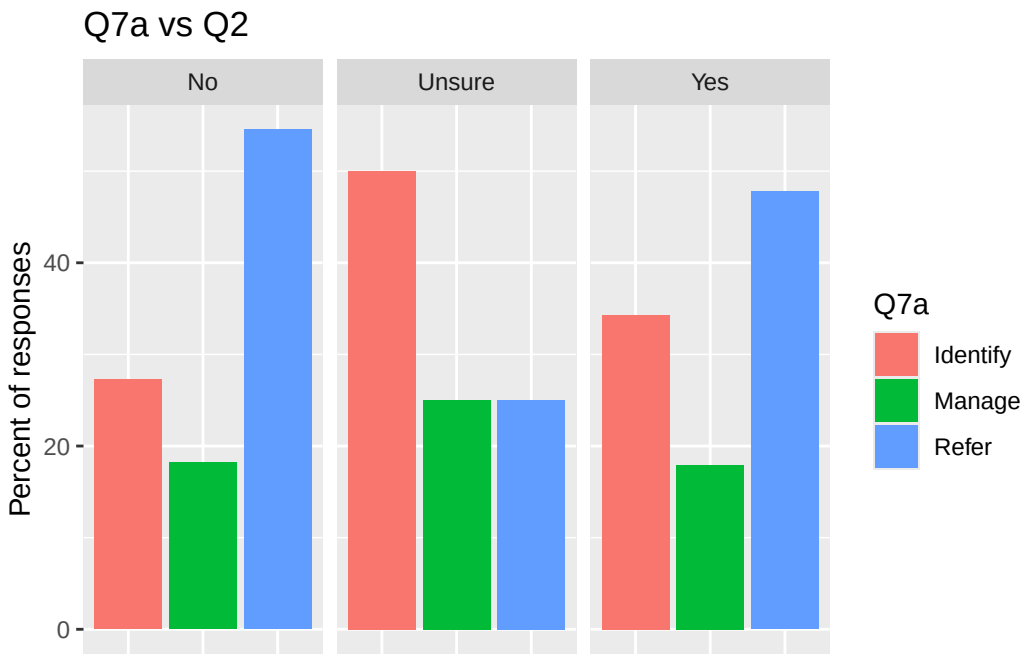
c) manage pre-surgical and post-surgical dental care for patients with cleft or…

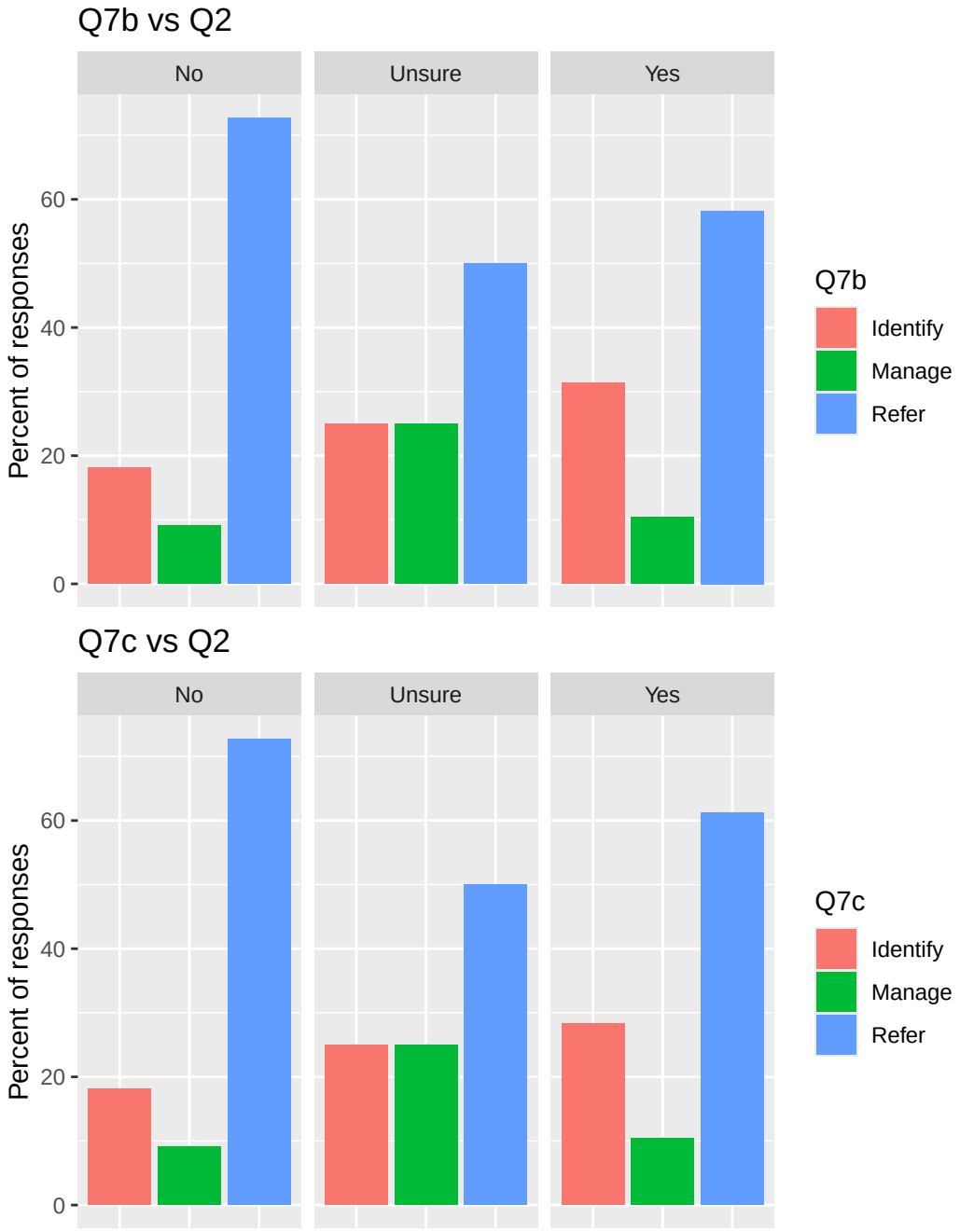

Q6 vs role
Director
Resident
Percent of responses
100
75
50
25
0
Q6
No
Unsure
Yes
Q7b vs role
Director
Resident
Percent of responses
60
40
20
0
Q7b
Identify
Manage
Refer

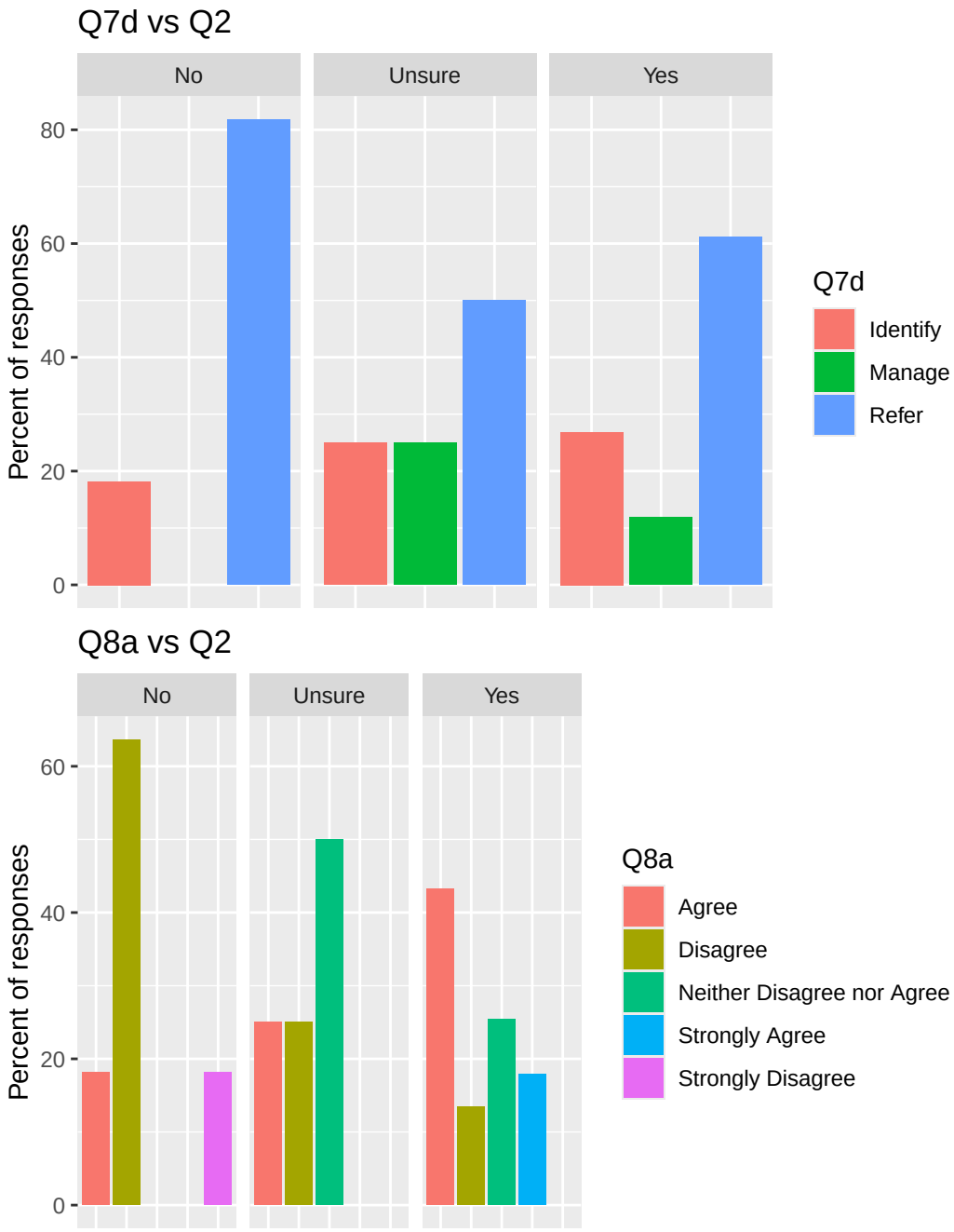

Q8a vs role
Director
Resident
Percent of responses
40
20
0
Q8a
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Strongly Disagree
Q8b vs role
Director
Resident
Percent of responses
40
20
0
Q8b
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree

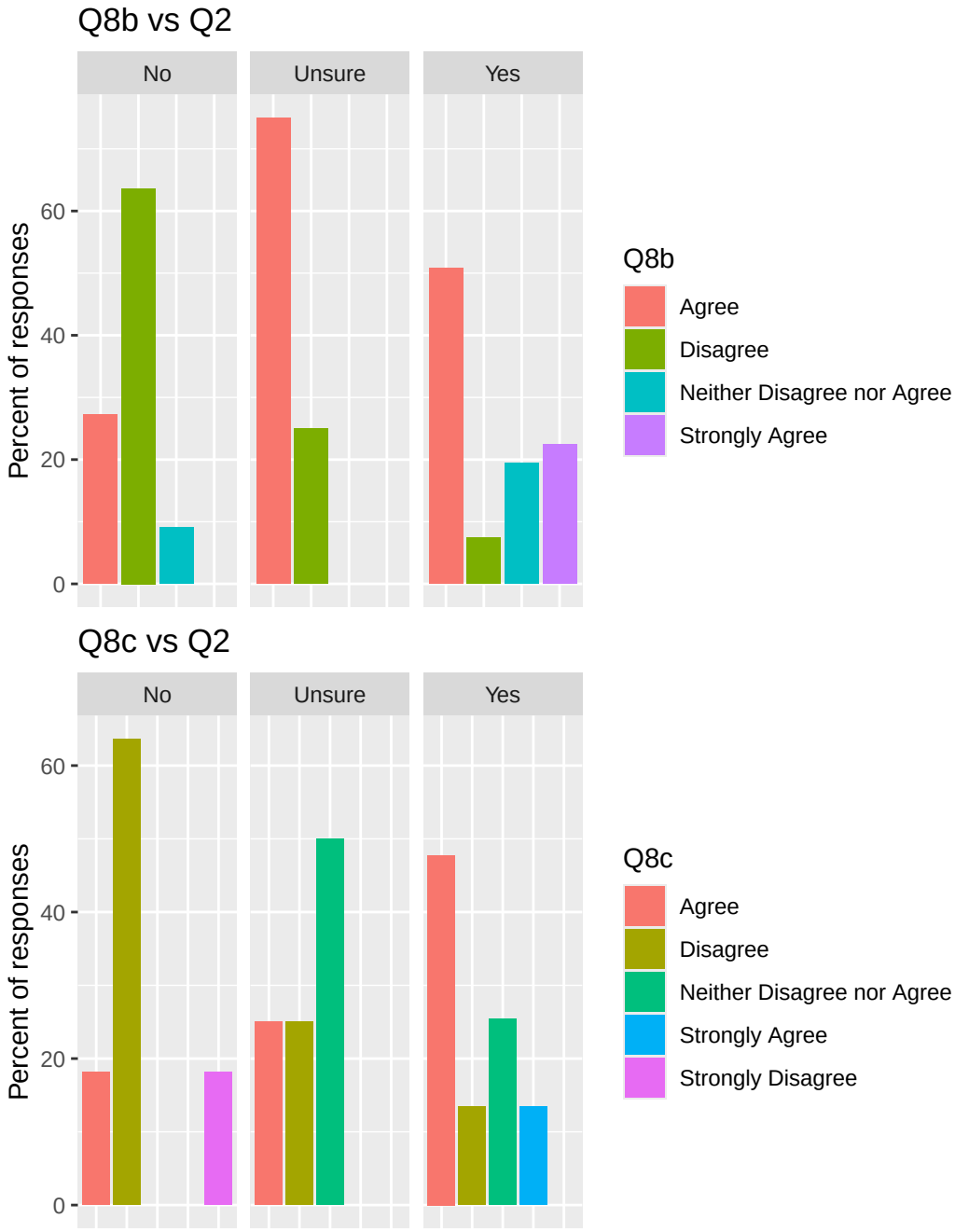


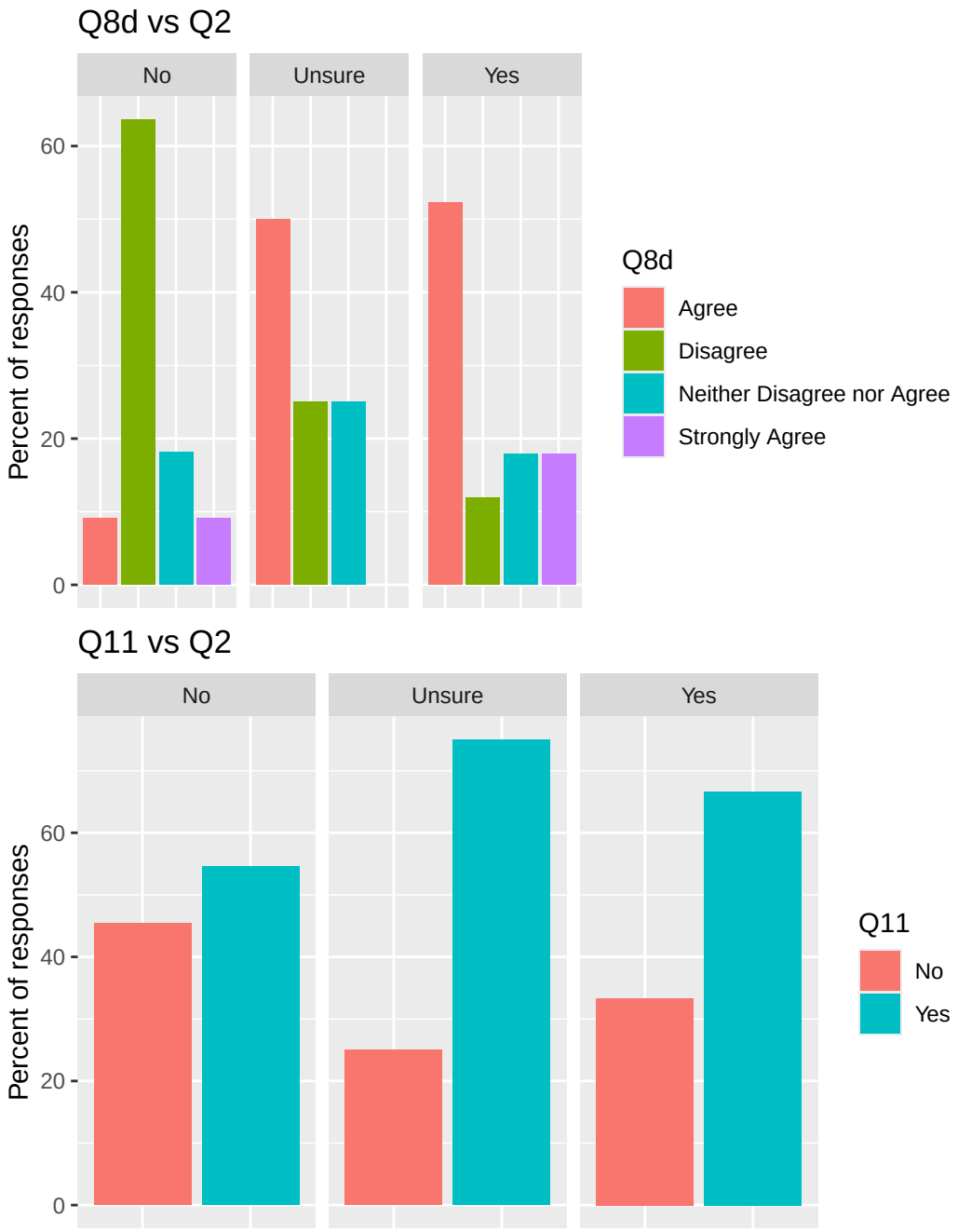
Appendix A i) Residents Visualization (all plots)

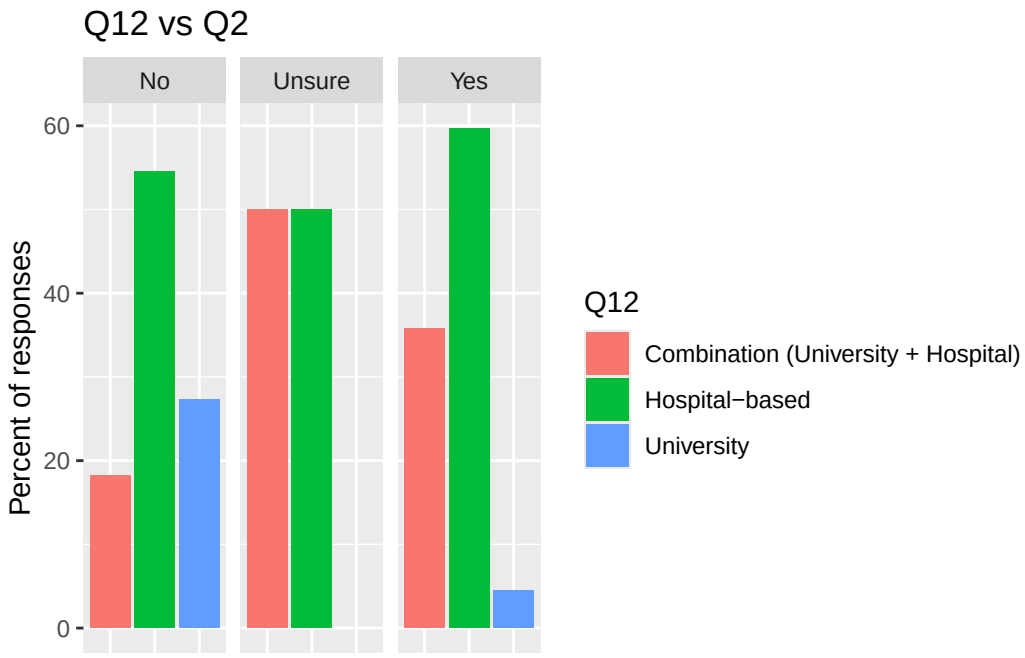



Q7b vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q7b
Identify
Manage
Refer
Q7c vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q7c
Identify
Manage
Refer

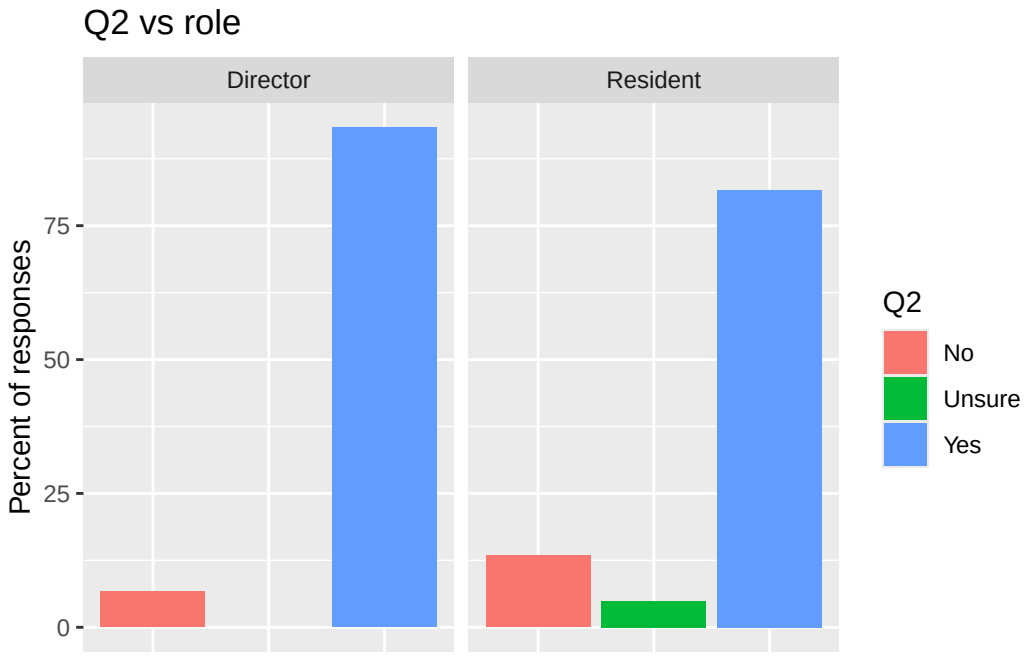

Q7d vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
80
Q7d
Identify
Manage
Refer
Q8a vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q8a
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Strongly Disagree

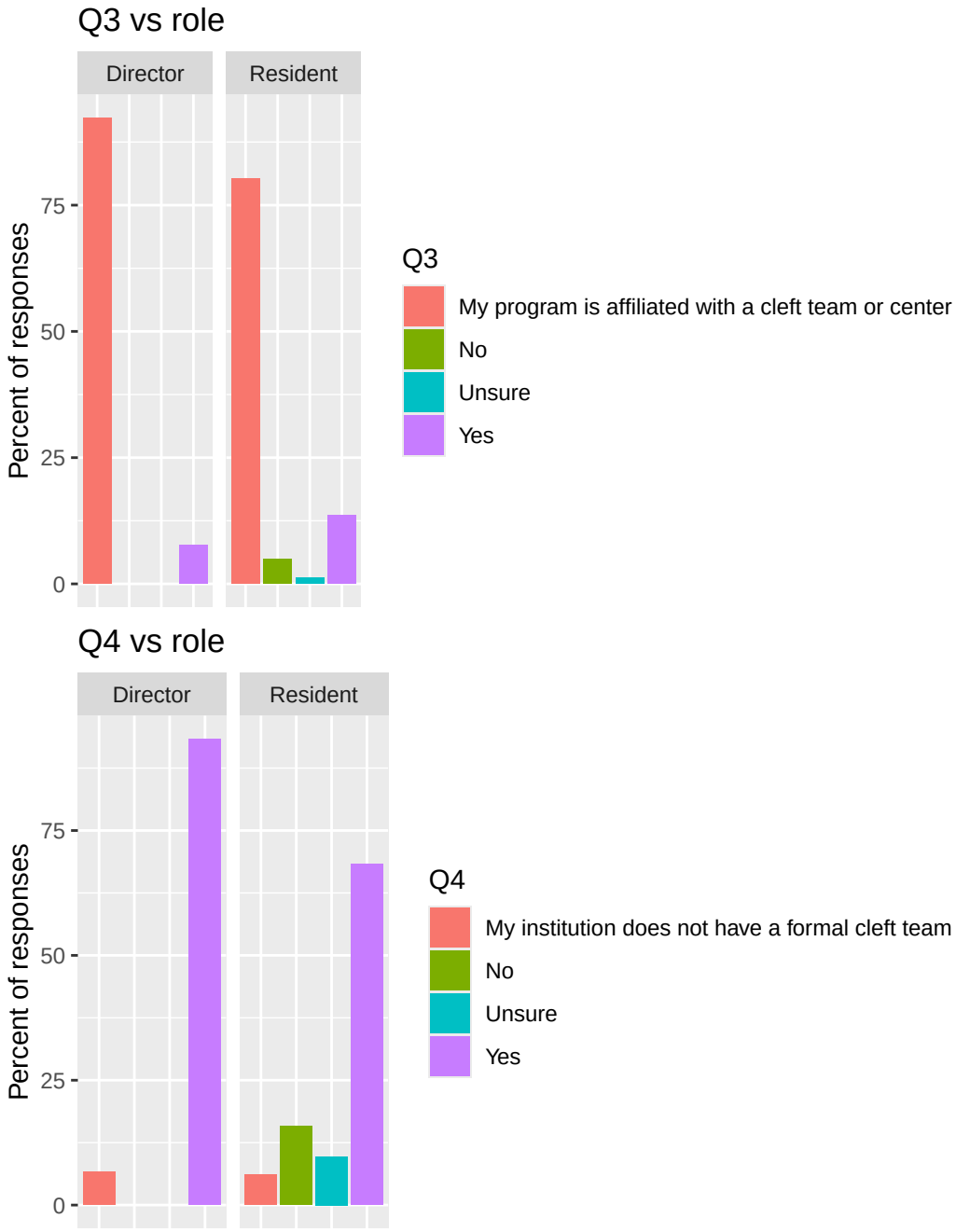

Q8b vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q8b
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Q8c vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q8c
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Strongly Disagree

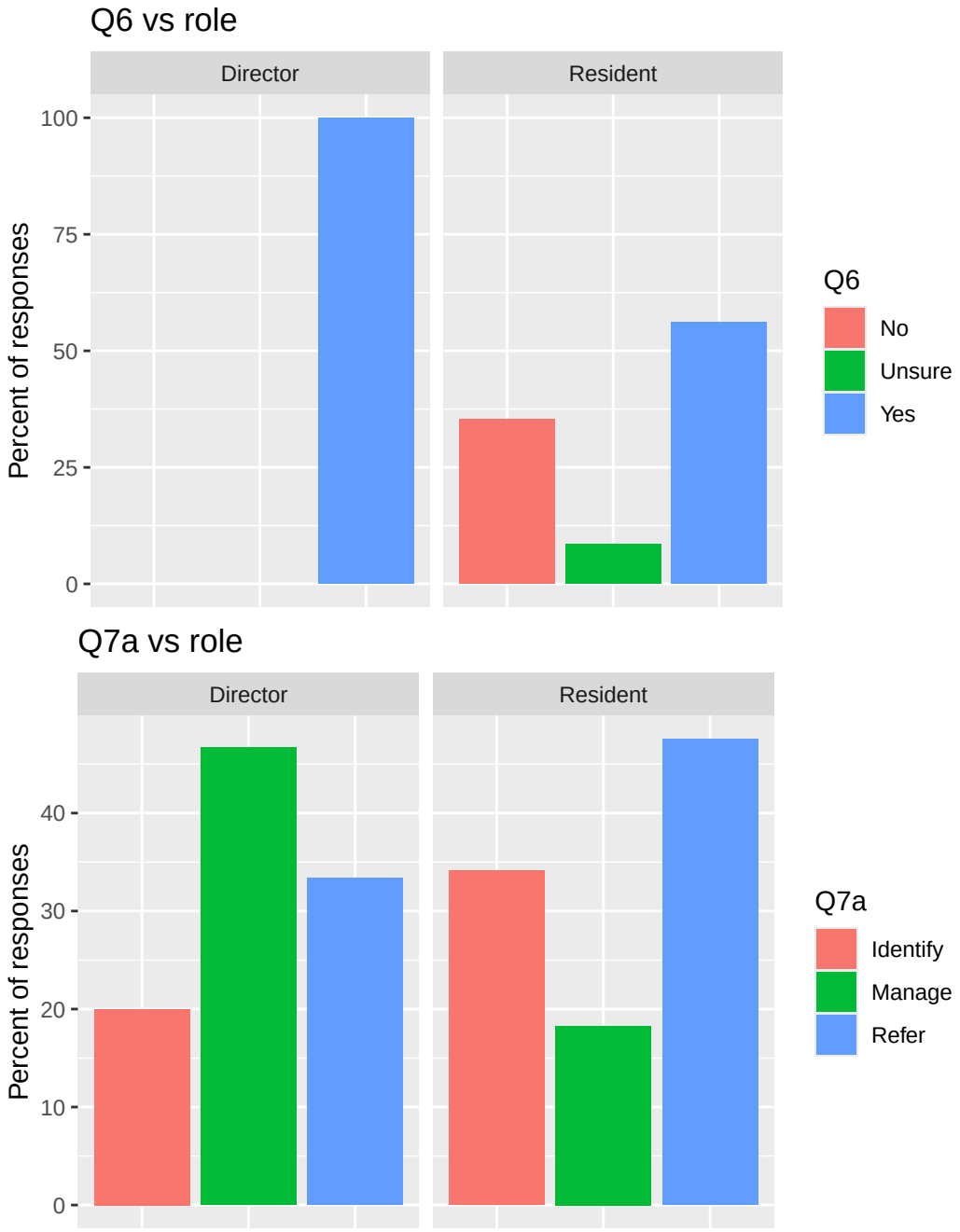

Q8d vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q8d
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Q11 vs Q2
No
Unsure
Yes
Percent of responses
0
20
40
60
Q11
No
Yes

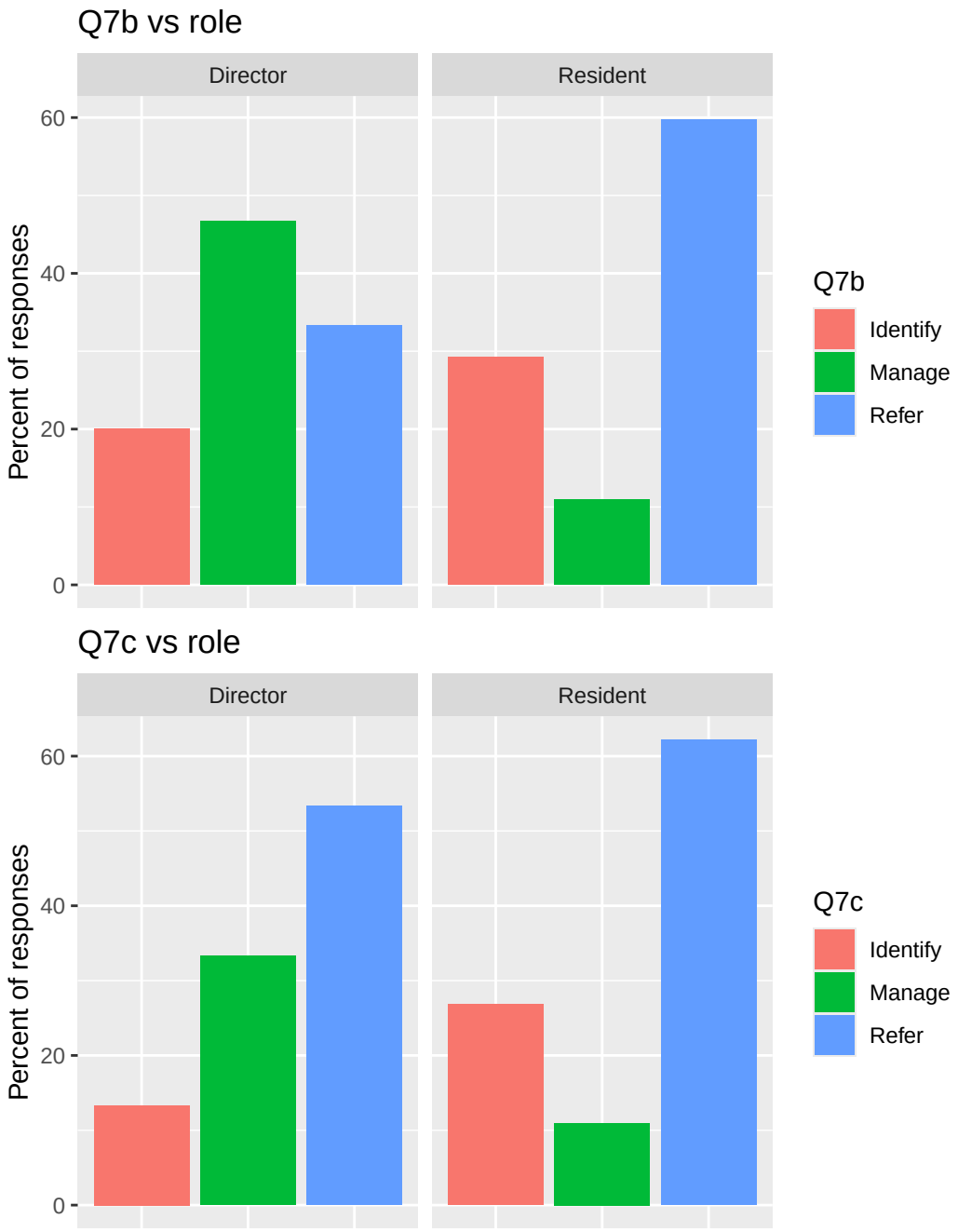


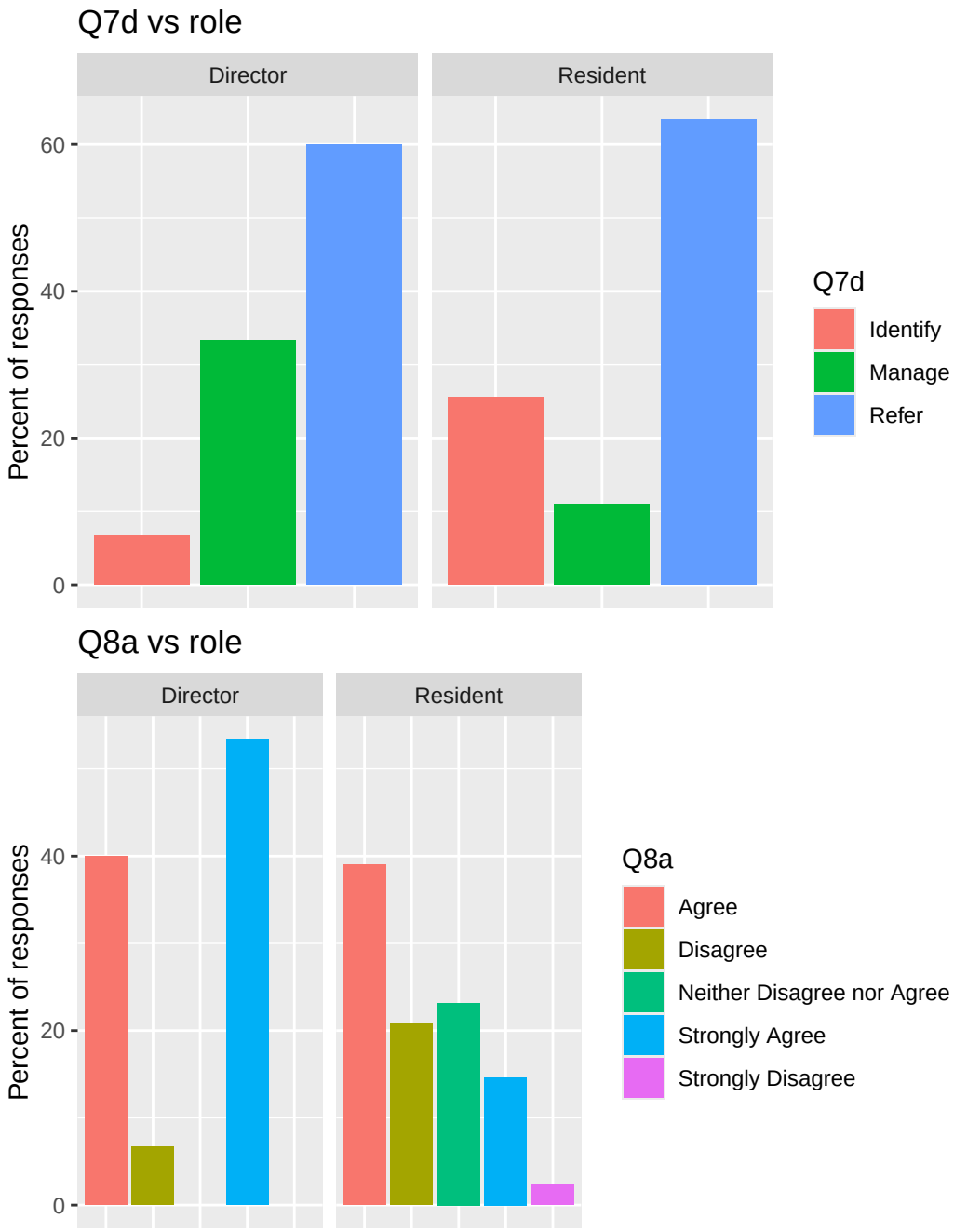
Appendix A ii) Cross-reference visualization (all plots)

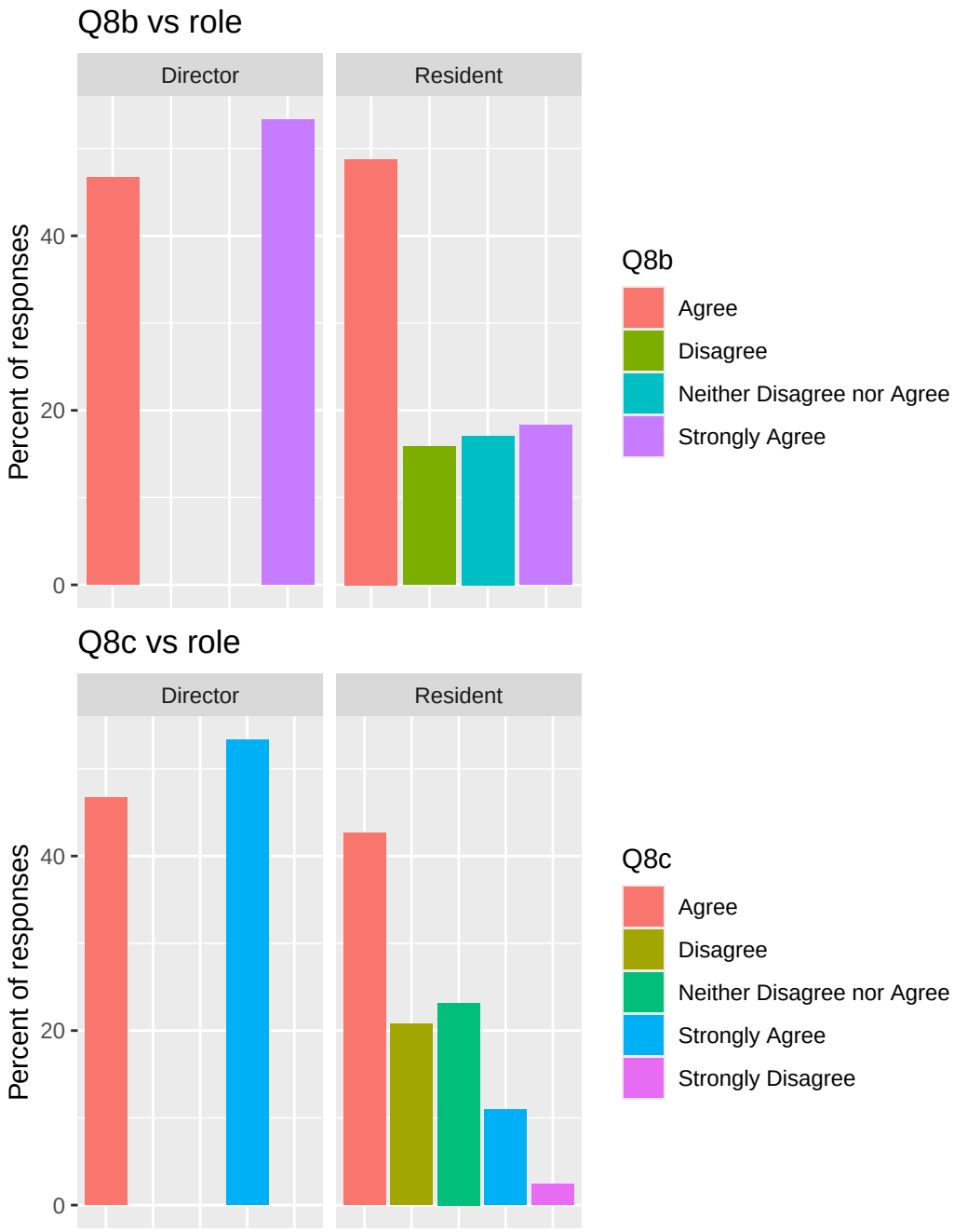


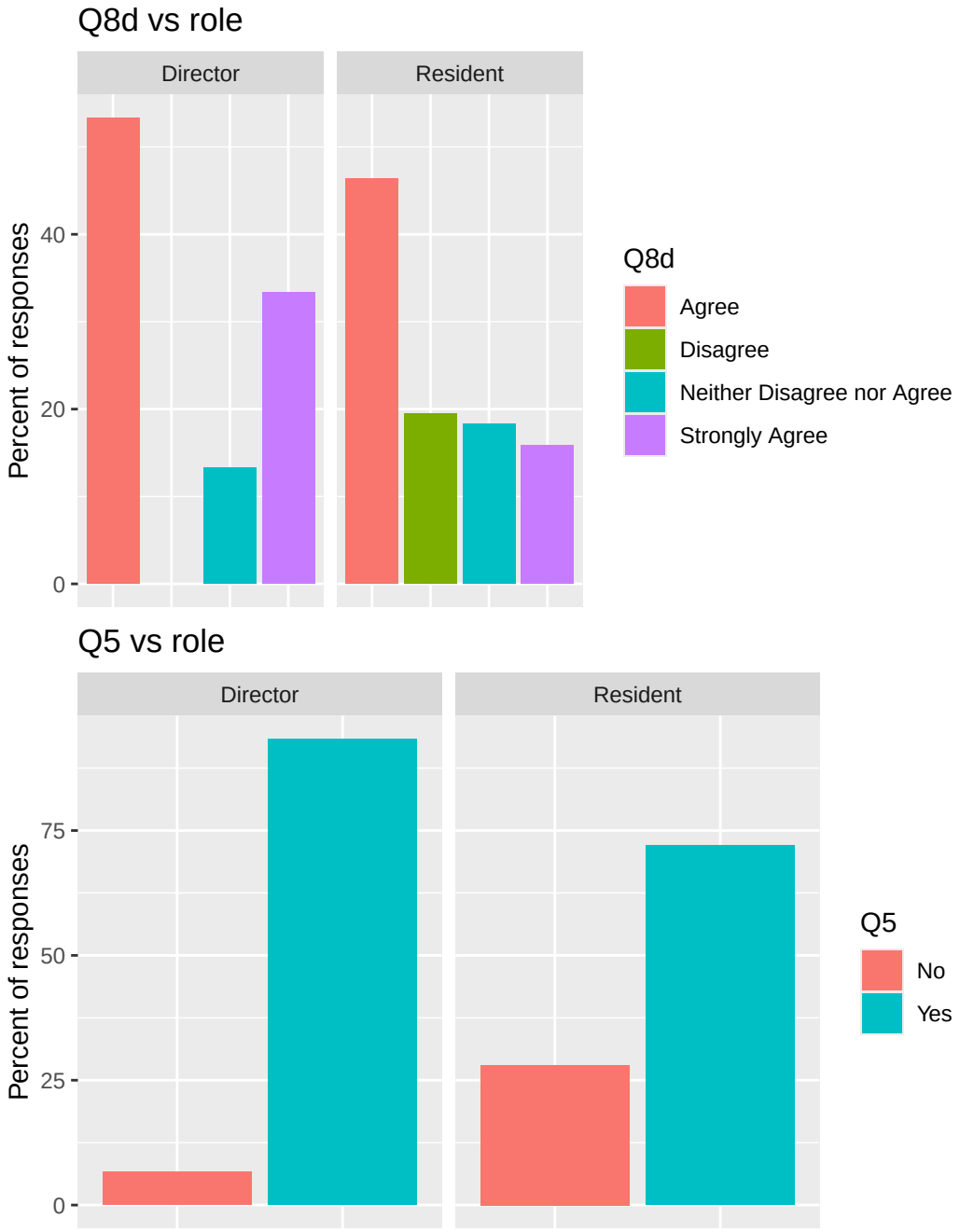

Q3 vs role
Director
Resident
Percent of responses
75
50
25
0
Q3
My program is affiliated with a cleft team or center
No
Unsure
Yes
Q4 vs role
Director
Resident
Percent of responses
75
50
25
0
Q4
My institution does not have a formal cleft team
No
Unsure
Yes

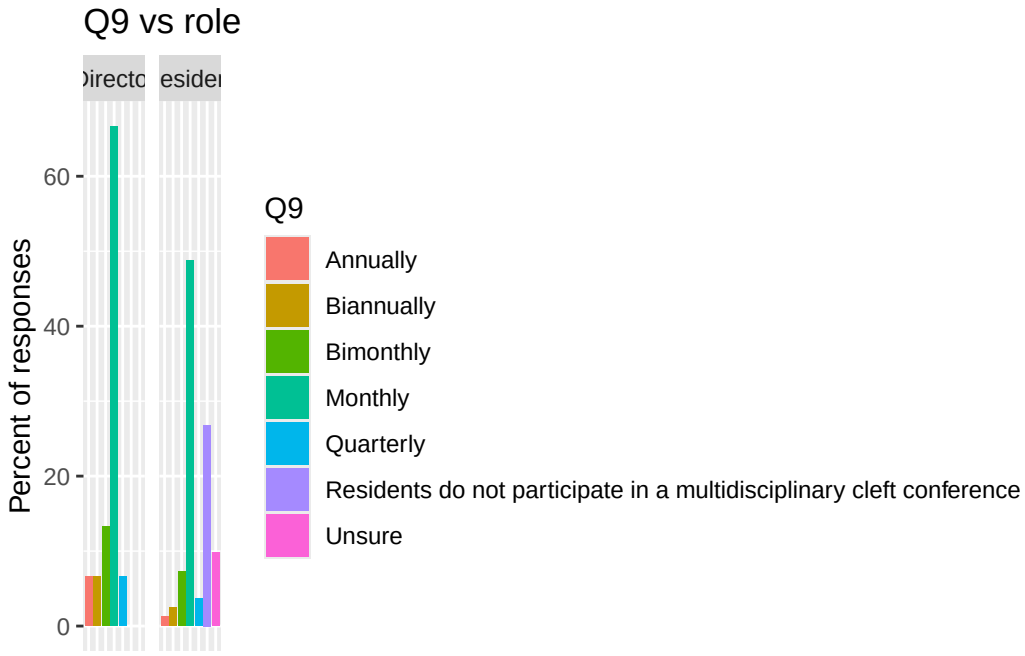

Q6 vs role
Director
Resident
Percent of responses
100
75
50
25
0
Q6
No
Unsure
Yes
Q7a vs role
Director
Resident
Percent of responses
40
30
20
10
0
Q7a
Identify
Manage
Refer

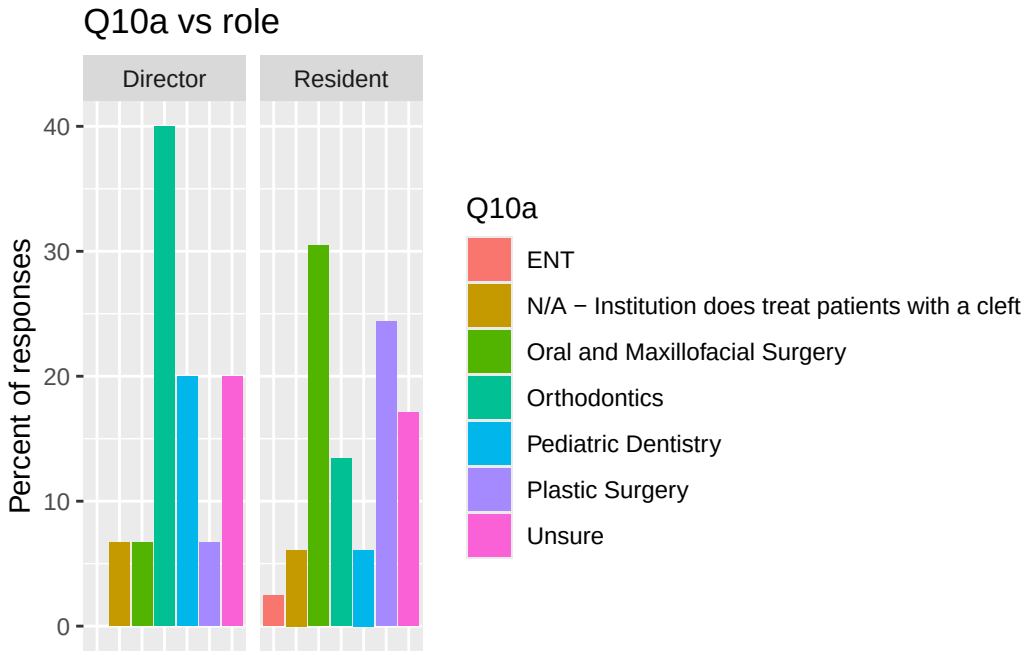

Q7b vs role
Director
Resident
Percent of responses
60
40
20
0
Q7b
Identify
Manage
Refer
Q7c vs role
Director
Resident
Percent of responses
60
40
20
0
Q7c
Identify
Manage
Refer


Q7d vs role
Director
Resident
Percent of responses
60
40
20
0
Q7d
Identify
Manage
Refer
Q8a vs role
Director
Resident
Percent of responses
40
20
0
Q8a
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Strongly Disagree

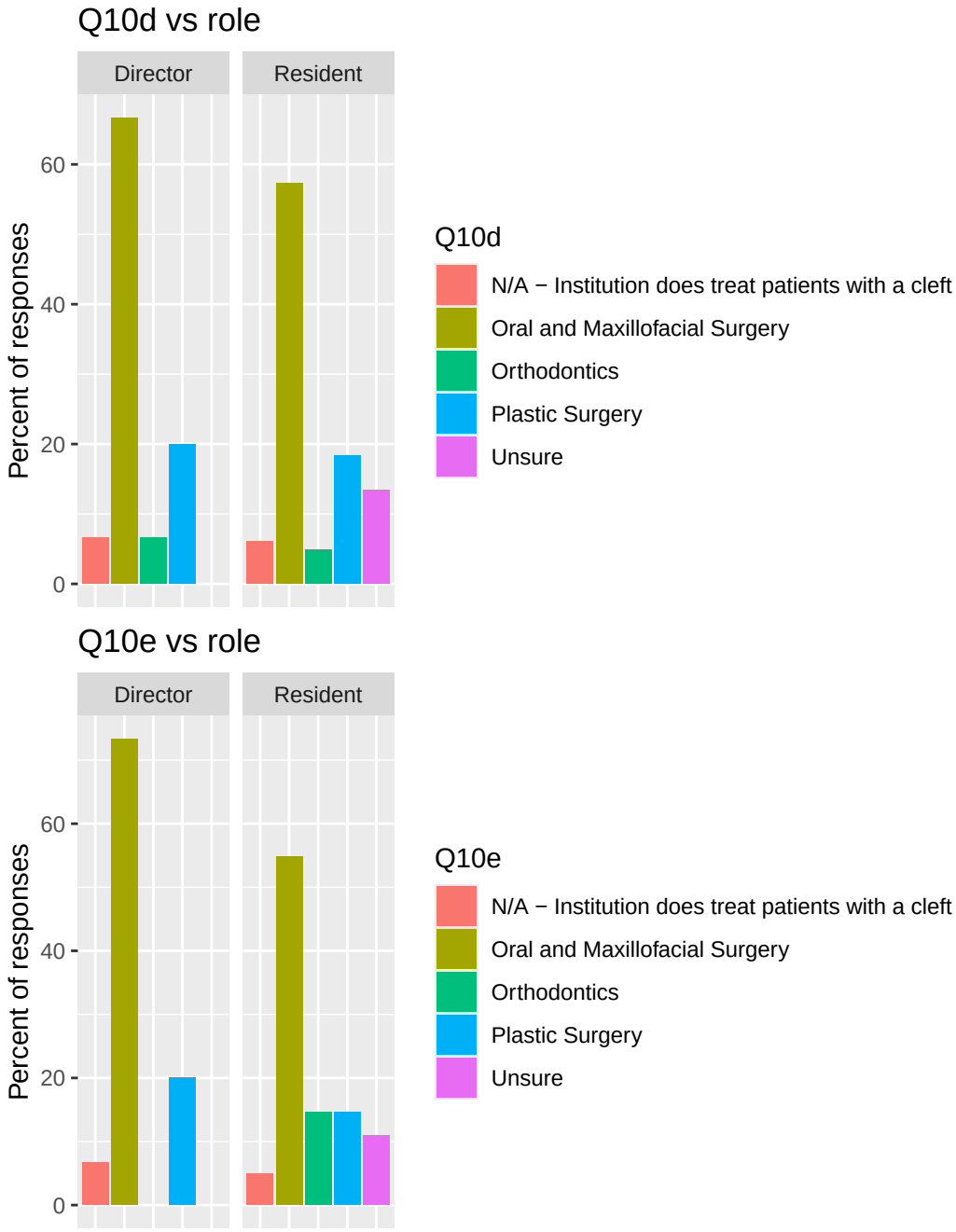

Q8b vs role
Director
Resident
Percent of responses
40
20
0
Q8b
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Q8c vs role
Director
Resident
Percent of responses
40
20
0
Q8c
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Strongly Disagree

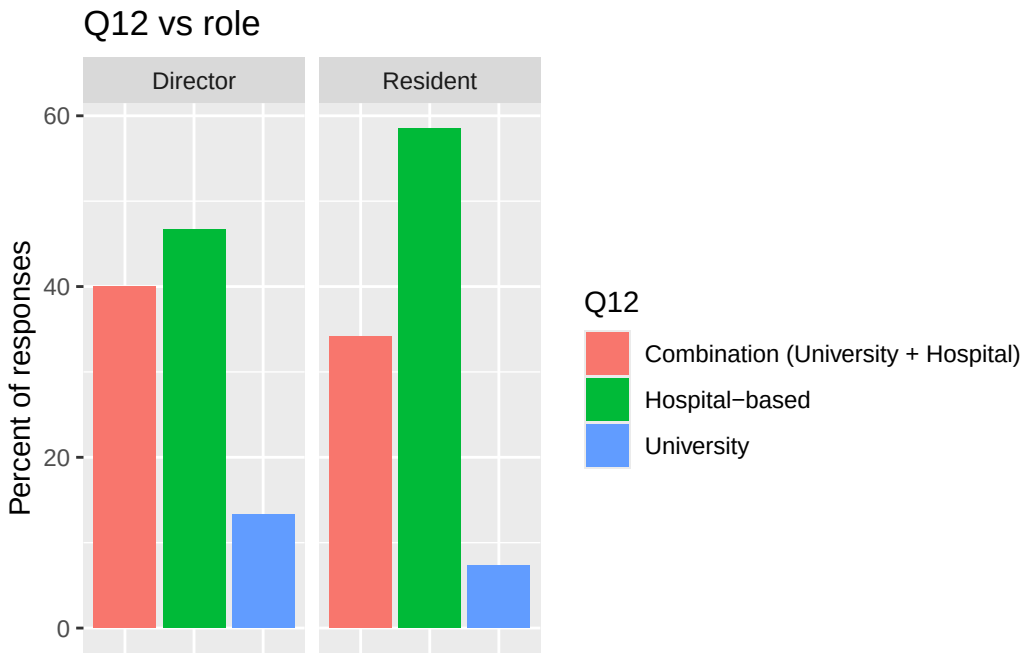

Q8d vs role
Director
Resident
Percent of responses
40
20
0
Q8d
Agree
Disagree
Neither Disagree nor Agree
Strongly Agree
Q5 vs role
Director
Resident
Percent of responses
75
50
25
0
Q5
No
Yes


Q9 vs role
Percent of responses
60
40
20
0
Q9
Annually
Biannually
Bimonthly
Monthly
Quarterly
Residents do not participate in a multidisciplinary cleft conference
Unsure


Q10a vs role
Director
Resident
Percent of responses
40
30
20
10
0
Q10a
ENT
N/A − Institution does treat patients with a cleft
Oral and Maxillofacial Surgery
Orthodontics
Pediatric Dentistry
Plastic Surgery
Unsure


Q10b vs role
Director
Resident
Percent of responses
60
40
20
0
Q10b
ENT
N/A − Institution does treat patients with a cleft
Oral and Maxillofacial Surgery
Plastic Surgery
Unsure
Q10c vs role
Director
Resident
Percent of responses
40
20
0
Q10c
N/A − Institution does treat patients with a cleft
Oral and Maxillofacial Surgery
Plastic Surgery
Unsure


Q10d vs role
Director
Resident
Percent of responses
60
40
20
0
Q10d
N/A − Institution does treat patients with a cleft
Oral and Maxillofacial Surgery
Orthodontics
Plastic Surgery
Unsure
Q10e vs role
Director
Resident
Percent of responses
60
40
20
0
Q10e
N/A − Institution does treat patients with a cleft
Oral and Maxillofacial Surgery
Orthodontics
Plastic Surgery
Unsure



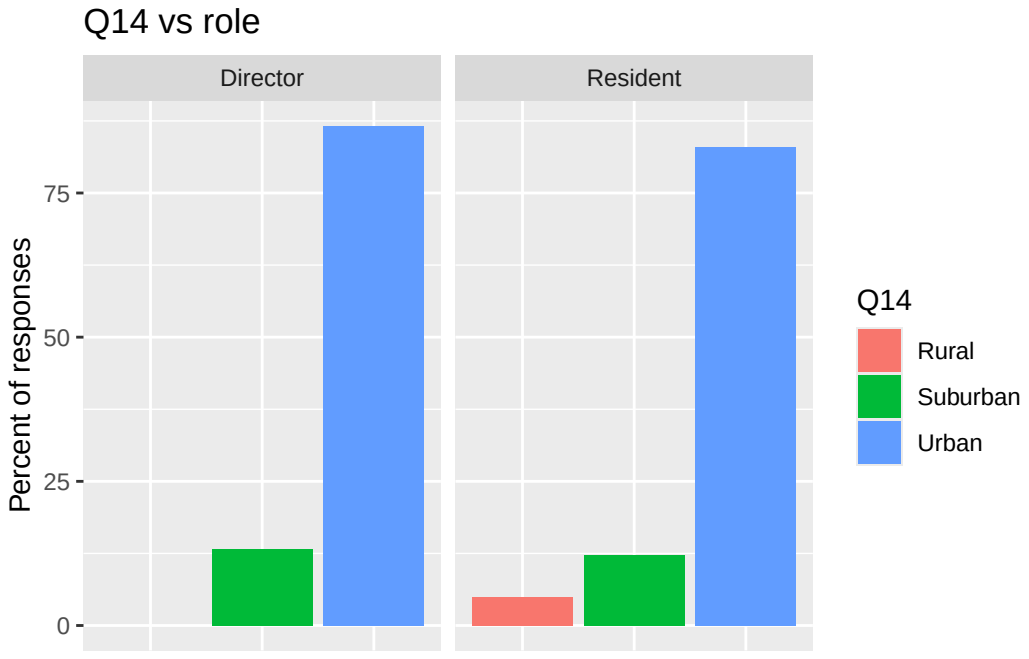
ı, Ohio, South Dakota, Wisconsin, and the Canadian provinces of Manitoba and Ontario)

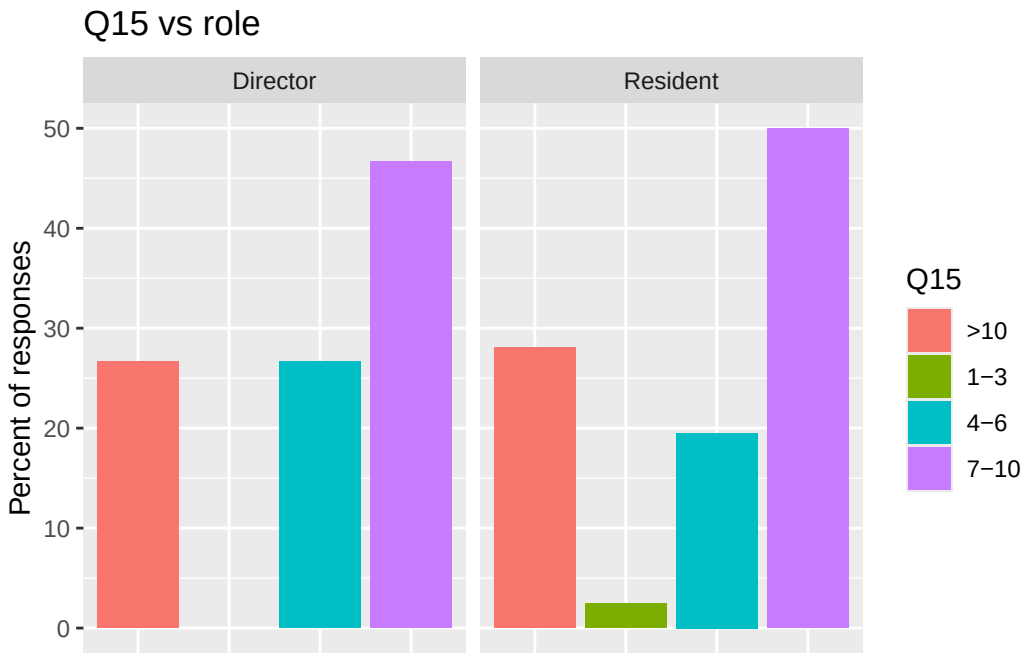
usetts, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, Vermont, and the Cana

ı Carolina, Tennessee, Virginia, West Virginia, the Commonwealth of Puerto Rico, and the US Virgin

and Mexico

Washington, Wyoming, and the Canadian provinces of Saskatchewan, Alberta, British Columbia, No





Appendix B: Code

```
library(tidyverse)
library(zoo)
library(stringr)
library(gt)
set.seed(1)

i_res_headers <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/CSV/res_survey.csv", n_max

questions <- as.character(i_res_headers[1,])

answer_choices <- as.character(i_res_headers[2,])

questions_filled <- na.locf(ifelse(questions == "" | is.na(questions), NA, questions))

clean_headers <- paste(questions_filled, answer_choices, sep = "_")

i_res_raw <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/CSV/res_survey.csv", skip = 2,
i_res <- i_res_raw %>%
  select(c(10:ncol(i_res_raw)))
```

```r
i_res <- i_res %>%
  mutate(
    across(
      everything(),
      ~ na_if(as.character(.), "")
    )
  )
i_res <- i_res %>%
  select(!"Which departments are primarily responsible for management of the following proce
i_res_q <- i_res %>%
  mutate(Q1 = do.call(coalesce, pick(1:2))) %>%
  mutate(Q2 = do.call(coalesce, pick(3:5))) %>%
  mutate(Q3 = do.call(coalesce, pick(6:9))) %>%
  mutate(Q4 = do.call(coalesce, pick(10:13))) %>%
  mutate(Q5 = do.call(coalesce, pick(14:15))) %>%
  mutate(Q6 = do.call(coalesce, pick(16:18))) %>%
  mutate(Q7a = do.call(coalesce, pick(19:21))) %>%
  mutate(Q7b = do.call(coalesce, pick(22:24))) %>%
  mutate(Q7c = do.call(coalesce, pick(25:27))) %>%
  mutate(Q7d = do.call(coalesce, pick(28:30))) %>%
  mutate(Q8a = do.call(coalesce, pick(31:35))) %>%
  mutate(Q8b = do.call(coalesce, pick(36:40))) %>%
  mutate(Q8c = do.call(coalesce, pick(41:45))) %>%
  mutate(Q8d = do.call(coalesce, pick(46:50))) %>%
  mutate(Q9 = do.call(coalesce, pick(51:57))) %>%
  mutate(Q10a = do.call(coalesce, pick(58:64))) %>%
  mutate(Q10b = do.call(coalesce, pick(65:71))) %>%
  mutate(Q10c = do.call(coalesce, pick(72:78))) %>%
  mutate(Q10d = do.call(coalesce, pick(79:85))) %>%
  mutate(Q10e = do.call(coalesce, pick(86:92))) %>%
  mutate(Q11 = do.call(coalesce, pick(93:94))) %>%
  mutate(Q12 = do.call(coalesce, pick(95:97))) %>%
  mutate(Q13 = do.call(coalesce, pick(98:102))) %>%
  mutate(Q14 = do.call(coalesce, pick(103:105))) %>%
  mutate(Q15 = do.call(coalesce, pick(106:109)))

i_res_q$role <- "Resident"

i_res_coalesced <- i_res_q %>%
  select(110:135)
i_dir_headers <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/i_directors.csv", n_max = 1
```

```r
questions_dir <- as.character(i_dir_headers[1,])

answer_choices_dir <- as.character(i_dir_headers[2,])

questions_filled_dir <- na.locf(ifelse(questions_dir == "" | is.na(questions_dir), NA, questi

clean_headers_dir <- paste(questions_filled_dir, answer_choices_dir, sep = "_")

i_dir_raw <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/i_directors.csv", skip = 2, col
i_dir <- i_dir_raw %>%
  select(c(10:ncol(i_dir_raw)))

i_dir <- i_dir %>%
  mutate(
    across(
      everything(),
      ~ na_if(as.character(.), "")
    )
  )
i_dir <- i_dir %>%
  select(!"Which departments are primarily responsible for management of the following proce
i_dir_q <- i_dir %>%
  mutate(Q1 = do.call(coalesce, pick(1:2))) %>%
  mutate(Q2 = do.call(coalesce, pick(3:5))) %>%
  mutate(Q3 = do.call(coalesce, pick(6:9))) %>%
  mutate(Q4 = do.call(coalesce, pick(10:13))) %>%
  # mutate(Q5 = do.call(coalesce, pick(14:25))) %>%
  mutate(Q6 = do.call(coalesce, pick(26:30))) %>%
  # mutate(Q7a = do.call(coalesce, pick(31:37))) %>%
  mutate(Q8 = do.call(coalesce, pick(38:42))) %>%
  mutate(Q9 = do.call(coalesce, pick(43:45))) %>%
  mutate(Q10a = do.call(coalesce, pick(46:48))) %>%
  mutate(Q10b = do.call(coalesce, pick(49:51))) %>%
  mutate(Q10c = do.call(coalesce, pick(52:54))) %>%
  mutate(Q10d = do.call(coalesce, pick(55:57))) %>%
  mutate(Q11a = do.call(coalesce, pick(58:62))) %>%
  mutate(Q11b = do.call(coalesce, pick(63:67))) %>%
  mutate(Q11c = do.call(coalesce, pick(68:72))) %>%
  mutate(Q11d = do.call(coalesce, pick(73:77))) %>%
  mutate(Q12 = do.call(coalesce, pick(78:79))) %>%
  mutate(Q13 = do.call(coalesce, pick(80:86))) %>%
  mutate(Q14 = do.call(coalesce, pick(87:90))) %>%
```

```
  mutate(Q15 = do.call(coalesce, pick(91:94))) %>%
  mutate(Q16a = do.call(coalesce, pick(95:101))) %>%
  mutate(Q16b = do.call(coalesce, pick(102:108))) %>%
  mutate(Q16c = do.call(coalesce, pick(109:115))) %>%
  mutate(Q16d = do.call(coalesce, pick(116:122))) %>%
  mutate(Q16e = do.call(coalesce, pick(123:129))) %>%
  mutate(Q17 = do.call(coalesce, pick(130:131))) %>%
  mutate(Q18 = do.call(coalesce, pick(132:134))) %>%
  mutate(Q19 = do.call(coalesce, pick(135:139))) %>%
  mutate(Q20 = do.call(coalesce, pick(140:142))) %>%
  mutate(Q21 = do.call(coalesce, pick(143:146)))

i_dir_coalesced_partial <- i_dir_q %>%
  select(147:150, 14:25, 151, 31:37, 152:175)

i_dir_coalesced <- i_dir_coalesced_partial %>%
  mutate(Q5 = pmap_chr(select(., 5:16), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  mutate(Q7 = pmap_chr(select(., 18:24), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  select(1:4, 49, 17, 50, 25:48)

i_dir_coalesced$role <- "Director"
i_dir_coalesced <- i_dir_coalesced %>%
  mutate(Q21_new = case_when(
    Q21 == "3-Jan" ~ "1-3",
    Q21 == "6-Apr" ~ "4-6",
    Q21 == "10-Jul" ~ "7-10",
    Q21 == ">10" ~ ">10"
  ))
# 2 with 7 and 8
# 2 with 11
# 2 with 11 and 12
question1s <- c("Q7a", "Q7b", "Q7c", "Q7d", "Q8a", "Q8b", "Q8c", "Q8d", "Q11", "Q12")

question2s <- c(rep("Q2", 10))

question_dic_res <- data.frame(question1s, question2s)
validation <- function(question1, question2, data){
  df <- data %>%
    transmute(
      response1 = .data[[question1]],
      response2 = .data[[question2]]
    ) %>%
```

```
    drop_na()

  comparison_res <- df %>%
    group_by(response1, response2) %>%
    summarize(counts = n(), .groups = "drop") %>%
    group_by(response2) %>%
    mutate(percent = counts / sum(counts) * 100) %>%
    ungroup()

  table_comparison <- table(df$response2, df$response1)
  #table_hours
  test <- suppressWarnings(chisq.test(table_comparison))

  if (mean(test$expected <= 5) < 0.8 | any(test$expected < 1)){
    fish_test <- fisher.test(table_comparison, simulate.p.value = TRUE)
    if (fish_test$p.value <= 0.05){
      significant <- TRUE
    }
    else {
      significant <- FALSE
    }
    return(c("question1", "question2", round(fish_test$p.value, digits = 4), significant, "fi
  }
  else{
    if (test$p.value <= 0.05){
      significant <- TRUE
    }
    else {
      significant <- FALSE
    }
  }
  return(c("question1", "question2", round(test$p.value, digits = 4), significant, "chisq"))
}
res_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(nrow(question_dic_res))) {
```

```
  res_results[i, ] <- validation(
    question1 = question_dic_res$question1s[i],
    question2 = question_dic_res$question2s[i],
    data = i_res_coalesced
  )
}

res_results <- res_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = question_dic_res$question1s,
    Question2 = question_dic_res$question2s
  ) %>%
  arrange(Adj_P)

library(modelsummary)
cleft_program <- i_res_coalesced %>%
  filter(Q2 != "Unsure") %>%
  mutate(cleft_response = as.factor(Q11),
         program = as.factor(Q12),
         cleft_center = as.factor(Q2)) %>%
  select(cleft_response, cleft_center, program) %>%
  drop_na()
cleft_glm <- glm(cleft_response ~ cleft_center + program, family = "binomial", data = cleft_p
cleft_null <- glm(cleft_response ~ 1, family = "binomial", data = cleft_program)
#logistic regression
#use model summary
# we notice signficance when comparing the model with program and cleft centers to the null m
common_qs_res <- c("Q2", "Q3", "Q4", "Q6", "Q7a", "Q7b", "Q7c", "Q7d","Q8a","Q8b", "Q8c", "Q8
                   "role")
common_qs_dir <- c("Q2", "Q3", "Q4", "Q9", "Q10a", "Q10b", "Q10c", "Q10d", "Q11a","Q11b","Q11
                  "Q21_new",   "role")
q_dic <- data.frame(common_qs_res, common_qs_dir)

res_common <- i_res_coalesced %>%
  select(all_of(common_qs_res))

dir_common <- i_dir_coalesced %>%
  select(all_of(common_qs_dir)) %>%
  setNames(common_qs_res)
res_dir <- bind_rows(
```

```
  res_common,
  dir_common
)
res_dir_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  Test = character()
)

for (i in seq_len(ncol(res_dir)-1)) {

  res_dir_results[i, ] <- validation(
    question1 = colnames(res_dir)[i],
    question2 = colnames(res_dir)[ncol(res_dir)],
    data = res_dir
  )
}

res_dir_results <- res_dir_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = colnames(res_dir)[1:(ncol(res_dir) -1)],
    Question2 = "role"
  ) %>% arrange(Adj_P)
stack_bar <- function(q_reference, q_group, data){
  bar_df <- data %>%
    group_by(.data[[q_reference]], .data[[q_group]]) %>%
    summarize(counts = n(), .groups = "drop") %>%
    drop_na() %>%
    group_by(.data[[q_group]]) %>%
    mutate(percent = counts / sum(counts) * 100) %>%
    ungroup()

  p <- ggplot(bar_df, aes(x = .data[[q_reference]], y = percent, fill = .data[[q_reference]]
    geom_col() +
    facet_wrap(vars(.data[[q_group]])) +
    labs(
      title = paste(q_reference, "vs", q_group, sep = " "),
      x = NULL,
```

```
      y = "Percent of responses"
    ) +
    theme(
      axis.text.x = element_blank(),
      axis.ticks.x = element_blank()
    )
  return(p)
}
plots_res <- pmap(question_dic_res,
                  function(question1s, question2s){
                    stack_bar(
                      q_reference = question1s,
                      q_group = question2s,
                      data = i_res_coalesced
                    )
                  })


question_dic_res_dir <- data.frame(question1s_res_dir = colnames(res_dir)
                                   , question2s_res_dir = rep("role", ncol(res_dir)))

question_dic_res_dir <- question_dic_res_dir %>%
  filter(question1s_res_dir != "role")
plots_comp <- pmap(question_dic_res_dir,
                   function(question1s_res_dir, question2s_res_dir){
                     stack_bar(
                       q_reference = question1s_res_dir,
                       q_group = question2s_res_dir,
                       data = res_dir
                     )
                   })
res_results

res_dir_results
sig_res_plots <- plots_res[c(5, 6, 7, 8)]
walk(sig_res_plots, print)
modelsummary(cleft_glm)
plots_comp_sig <- plots_comp[c(4, 6, 9, 10, 11)]
walk(plots_comp_sig, print)
walk(plots_res, print)
walk(plots_comp, print)
```